

LANGZHONG, China

WMO: 573060

Lat: 31.5846N

Lon: 105.9776E

Elev: 385

StdP: 96.78

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.1	2.1	-7.7	2.1	8.3	-4.2	2.8	8.5	4.3	6.9	3.7	7.2	0.8	320	0.311

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.4	35.8	25.5	34.4	25.2	33.1	24.8	27.1	32.6	26.6	31.6	26.0	30.6	1.7	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	22.1	29.8	25.2	21.4	29.2	24.7	20.7	28.7	88.9	32.8	86.2	31.7	83.4	30.7	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	4.3	3.5	3.0	DB	-0.7	37.5	1.3	1.4	-1.6	38.5	-2.4	39.3	-3.1	40.0	-4.0	41.0
(5)				WB	-1.9	27.9	1.4	0.9	-2.9	28.5	-3.7	29.1	-4.4	29.6	-5.4	30.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	6.7	9.1	13.5	18.6	22.3	25.2	27.8	27.7	23.0	18.2	13.2	8.1	(6)
(7)		DBStd	7.79	2.18	2.70	3.44	3.49	2.64	2.39	2.36	2.95	3.03	2.79	2.83	2.22	(7)
(8)		HDD10.0	229	104	44	7	1	0	0	0	0	0	6	67		(8)
(9)		HDD18.3	1327	361	259	154	38	3	0	0	2	37	155	318		(9)
(10)		CDD10.0	3083	2	19	115	259	382	455	552	548	390	253	102	7	(10)
(11)		CDD18.3	1138	0	0	4	46	127	205	293	290	141	32	1	0	(11)
(12)		CDH23.3	10277	0	0	27	320	874	1707	3207	3171	871	99	1	0	(12)
(13)	CDH26.7	4198	0	0	3	82	258	624	1431	1495	292	14	0	0	(13)	
(14)	Wind	WSAvg	1.1	0.9	1.0	1.2	1.2	1.3	1.1	1.1	1.1	0.9	0.9	0.9		(14)
(15)	Precipitation	PrecAvg	1039	12	14	33	68	116	107	204	174	184	80	33	12	(15)
(16)		PrecMax	1698	28	44	78	217	360	236	461	600	535	210	72	30	(16)
(17)		PrecMin	622	0	3	6	16	19	12	34	22	6	24	2	1	(17)
(18)		PrecStd	254	8	9	19	41	80	53	99	114	124	40	18	9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.4	19.7	26.3	31.5	33.1	34.6	37.1	37.7	35.2	28.5	22.5	15.8	(19)
MCWB			9.2	12.1	16.7	19.8	20.9	23.8	25.7	25.3	24.6	20.2	15.8	10.2	(20)	
2%		DB	12.9	16.8	23.4	28.9	31.2	32.9	35.5	36.2	32.3	26.0	20.6	13.7	(21)	
		MCWB	8.1	10.5	15.5	18.8	20.4	23.7	25.7	25.5	23.9	19.3	15.1	9.4	(22)	
5%		DB	11.3	15.0	21.5	26.9	29.5	31.6	34.1	34.8	29.9	24.1	18.6	12.4	(23)	
		MCWB	7.3	9.9	14.3	18.2	20.3	23.3	25.6	25.4	22.7	18.5	14.3	8.9	(24)	
10%		DB	10.1	13.4	19.4	24.8	27.9	30.3	32.8	33.3	27.9	22.4	17.1	11.3	(25)	
		MCWB	6.9	9.2	13.4	17.5	19.9	23.0	25.3	25.2	22.1	18.0	14.0	8.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	12.9	18.2	21.8	23.7	26.5	28.1	27.8	26.2	21.8	17.1	12.0	(27)
MCDWB			13.2	17.8	24.6	29.0	29.1	31.4	33.6	34.1	32.1	26.2	20.3	13.7	(28)	
2%		WB	8.8	11.5	16.2	20.4	22.6	25.5	27.3	27.1	25.0	20.4	16.2	10.7	(29)	
		MCDWB	11.6	15.1	21.9	25.9	27.5	29.8	32.4	32.9	29.9	23.9	18.8	12.5	(30)	
5%		WB	8.1	10.6	15.0	19.3	21.8	24.7	26.7	26.5	24.0	19.6	15.5	9.8	(31)	
		MCDWB	10.4	13.7	20.0	24.6	26.6	28.9	31.6	31.9	28.1	22.6	17.7	11.5	(32)	
10%		WB	7.4	9.8	14.0	18.2	20.9	24.1	26.2	26.0	23.1	18.7	14.7	9.1	(33)	
		MCDWB	9.5	12.7	18.3	23.4	25.7	28.2	30.7	31.0	26.4	21.5	16.5	10.7	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.3	6.0	7.6	8.5	8.3	7.3	7.4	8.0	6.2	5.6	5.4	4.8	(35)
MCDBR			8.0	9.4	11.2	12.3	11.7	10.4	9.9	10.5	9.9	8.9	8.4	6.9	(36)	
MCWBR			4.2	4.5	4.5	4.3	3.6	3.1	2.7	2.5	2.9	3.3	3.8	3.7	(37)	
5% WB		MCDBR	6.8	8.0	10.2	10.3	9.1	8.1	8.1	9.1	7.8	7.2	6.5	5.3	(38)	
		MCWBR	3.9	4.2	4.4	4.1	3.8	2.9	2.9	2.9	2.9	3.1	3.3	3.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.715	0.764	0.774	0.743	0.693	0.657	0.665	0.695	0.709	0.752	0.758	0.718	(40)	
(41)		taud	1.425	1.368	1.385	1.475	1.611	1.709	1.715	1.638	1.584	1.468	1.429	1.453	(41)	
(42)		Ebn at Noon	494	525	571	621	661	682	674	645	608	527	462	462	(42)	
(43)		Edh at Noon	263	304	318	300	264	238	236	252	256	270	259	244	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.77	2.10	3.10	4.04	4.34	4.25	4.73	4.48	3.11	2.35	1.85	1.61	(44)	
(45)		RadStd	0.23	0.38	0.37	0.51	0.50	0.51	0.78	0.68	0.42	0.40	0.26	0.25	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.29	+0.64	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	+0.28	N/A	-0.85	N/A	N/A	N/A	-36	-79	+97	+38

Nomenclature: See separate page